

1. COUNTING POLYGON MOSAICS

Theorem 1. *Let v be any cell product function. Define the 1×1 matrices $A_1 = \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$, $B_1 = \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \end{pmatrix}$, $C_1 = \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$, and $D_1 = \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \end{pmatrix}$, and for positive integers n , define the $2^n \times 2^n$ matrices* Note that in the expressions below like $v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} B_n$, the LaTeX should be updated to $\backslashleft($ and $\backslashright)$.

$$\begin{aligned} A_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} D_n \end{pmatrix} & B_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} D_n \end{pmatrix} \\ C_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} D_n \end{pmatrix} & D_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} D_n \end{pmatrix}. \end{aligned}$$

Then for all positive integers m and n , the upper left entry of A_n^m equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .

In the statement of Theorem 1, we evaluate v at each of the sixteen possible cells at least once. The order in which these numbers appear may look haphazard but should be clear from the proof of Proposition 2, which we will use to prove Theorem 1. Until we finish those proofs, v and the matrices A_n, B_n, C_n , and D_n are as in Theorem 1. Let $\beta_n(k)$ be the n digit binary representation of the nonnegative integer $k < 2^n$, separated out into individual digits (e.g., $\beta_3(5)$ is 1 0 1 rather than 101), with $\beta_0(0)$ being the empty string. This allows us, for example, to write the binary lattice $\begin{smallmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{smallmatrix}$ as $\begin{smallmatrix} \beta_4(6) \\ \beta_4(2) \end{smallmatrix}$ or as $\begin{smallmatrix} 0 & \beta_2(3) & 0 \\ 0 & \beta_2(1) & 0 \end{smallmatrix}$.

Proposition 2. *Let n be a positive integer. Numbering rows and columns beginning at 0, we have the following.*

- (1) $\begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$ equals the $2^n \times 2^n$ matrix whose (i, j) entry is $v \begin{pmatrix} \beta_{n+1}(2i) \\ \beta_{n+1}(2j) \end{pmatrix}$.
- (2) A_n equals the $2^{n-1} \times 2^{n-1}$ matrix whose (i, j) entry is $v \begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}$.

Proof. Notice that for all nonnegative integers $k < 2^n$, $\beta_{n+1}(2k) = \beta_n(k)$ 0 and that if $k < 2^{n-1}$ then $\beta_n(k) = 0 \beta_{n-1}(k)$. Putting these together, the first part of the theorem implies the second.

For all positive integers n , define X_n to be the $2^n \times 2^n$ matrix whose (i, j) entry is $v \begin{pmatrix} \beta_{n+1}(2i) \\ \beta_{n+1}(2j) \end{pmatrix}$, so that we must prove $X_n = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$ for all n . Since $\beta_2(0)$ is 0 0 and $\beta_2(2)$ is 1 0, the $n = 1$ case follows from the definition of A_1, B_1, C_1 , and D_1 . Continuing by induction, we assume $X_n = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$ and prove

$$X_{n+1} = \begin{pmatrix} A_{n+1} & B_{n+1} \\ C_{n+1} & D_{n+1} \end{pmatrix}.$$

Consider nonnegative integers $i, j < 2^{n+1}$. Because v is a cell product function, $v \begin{pmatrix} \beta_{n+2}(2i) \\ \beta_{n+2}(2j) \end{pmatrix}$ is equal to v evaluated at the leftmost cell of $\begin{smallmatrix} \beta_{n+2}(2i) \\ \beta_{n+2}(2j) \end{smallmatrix}$ times v evaluated at the $(2, n)$ binary lattice formed by deleting the leftmost column of $\begin{smallmatrix} \beta_{n+2}(2i) \\ \beta_{n+2}(2j) \end{smallmatrix}$. To evaluate X_{n+1} , we thus can evaluate v on certain cells and on certain $(2, n)$ binary lattices. These will appear, respectively, as the coefficients and submatrices that define $\begin{pmatrix} A_{n+1} & B_{n+1} \\ C_{n+1} & D_{n+1} \end{pmatrix}$ in the statement of Theorem 1.

The sequence $\beta_{n+2}(0), \beta_{n+2}(2), \dots, \beta_{n+2}(2^{n+1} - 2)$ has 2^{n+1} elements. Dividing it into four equal parts, in the first [resp. second, third, and fourth] part, the

initial two digits are 0 0 [resp. 0 1, 1 0, and 1 1]. Therefore, as we go down any column of X_{n+1} from the top row to the bottom row, the cell to be evaluated has upper row 0 0 [resp. 0 1, 1 0, and 1 1] in the first [resp. second, third, and fourth] quarter of the entries. Similarly, going across any row of X_{n+1} , left to right, the lower row of the cell is 0 0, 0 1, 1 0, and 1 1 in four equal parts. This is exactly what we see in the definition of $\begin{pmatrix} A_{n+1} & B_{n+1} \\ C_{n+1} & D_{n+1} \end{pmatrix}$.

In $\begin{pmatrix} A_{n+1} & B_{n+1} \\ C_{n+1} & D_{n+1} \end{pmatrix}$, each quadrant, by definition includes submatrices that are $\begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$, which by our induction hypothesis equals X_n . Therefore, to complete our proposition we must show that in each quadrant of X_{n+1} , evaluating just the $(2, n)$ binary lattices gives us X_n . To see this, consider taking the sequence $\beta_{n+2}(0)$, $\beta_{n+2}(2)$, $\dots$, $\beta_{n+2}(2^{n+2} - 2)$ and deleting first digit of each string. This gives two copies of the sequence $\beta_{n+1}(0)$, $\beta_{n+1}(2)$, $\dots$, $\beta_{n+1}(2^{n+1} - 2)$. Therefore, within any column of X_n from the top row to the bottom row, the upper row of the $(2, n)$ lattice used goes twice through the strings $\beta_{n+1}(0)$, $\beta_{n+1}(2)$, $\dots$, $\beta_{n+1}(2^{n+1} - 2)$. Also, within any row of X_n from left to right, the lower row of the $(2, n)$ lattice also goes through these strings twice. Therefore, evaluating v at each of these lattices in each quadrant of X_{n+1} equals X_n , as desired. $\square$